

PRODUCT REQUIREMENTS DOCUMENT

Market Brain Grid

MBG

Multi-Agent Research & Trading Desk for IDX + Crypto
Powered by Hermes Agent · Debate · Shapley Credit · Human Review

Version 1.2 — Full Implementation Spec with Embedded Hermes Skills

Project Name	Market Brain Grid (MBG)
Version	1.2
Runtime	Hermes Agent (Nous Research)
Markets	Indonesia IDX/BEI + Global Crypto (CEX/DEX)
Core Pattern	Multi-agent debate + Shapley credit + safety gates
Execution Policy	Paper-first · Live only with human confirmation · Kill-switch
Date	2026-09-08

1. Executive Summary

Market Brain Grid (MBG) is a multi-agent autonomous research and supervised-trading system focused on the Indonesian equity market (IDX/BEI) and global cryptocurrency markets. It runs on the Hermes Agent runtime and is packaged as a set of reusable skills. MBG produces daily analysis, ranked recommendations, structured trade plans, and backtests. Every material recommendation ends with human review. Live capital is never touched without explicit confirmation and is protected by hard risk limits and a kill-switch.

Architecture draws from TradingAgents (role specialization + Bull/Bear debate), AutoHedge (risk-first pipeline), Vibe-Trading (NL research + backtest), Moss (strategy evolution), and Astra-style safety (FACTS vs OPINION, NEVER invent numbers, AWAITING HUMAN REVIEW). Credit assignment uses online Shapley values so the system can learn which agents and debate modes actually contribute to outcomes.

1.1 Goals

- Daily high-quality analysis and ranked recommendations for IDX stocks and major crypto pairs.
- Structured, auditable Trade Plans (entry, stop, targets, size, invalidation, confidence) with visible risk math.
- Multi-mode debate (adversarial, gated, regime-conditional, contest) with Research Manager synthesis.
- Online Shapley credit assignment that re-weights agents/coalitions from realized outcomes.
- Backtesting with point-in-time data and realistic costs.
- Deployable on laptop, Firebase, Supabase, or VPS via Hermes.
- Strict safety: paper-first, human gate for live, kill-switch, full audit log.

1.2 Non-Goals (v1)

- Fully unsupervised live capital deployment.
- High-frequency / sub-second market making.
- Tax, accounting, or regulatory filing automation.

2. Universal Safety Rules (Mandatory in Every Skill)

These rules are non-negotiable and must appear in every MBG skill prompt and every research/plan output.

1. Never invent a price, number, date, headline, or metric. If data is missing: **"NOT PROVIDED — verify manually"**.
2. Keep **FACTS** (with source + timestamp where available) strictly separate from **OPINION**.
3. No profit promises, no guaranteed returns, no buy/sell orders. Language stays research-oriented.
4. Every Trade Plan ends with: **STATUS: AWAITING HUMAN REVIEW** — nothing is executed until a human approves.
5. Position sizing must show the arithmetic: $(\text{account} \times \text{risk} \%) \div (\text{entry} - \text{stop}) = \text{shares}$.
6. If Risk/Reward is under ~2:1 at the proposed entry, the plan must say it is not worth taking at this price.
7. Paper mode is default. Live mode requires explicit human confirmation and is subject to kill-switch and mandate limits.
8. Full audit log of every agent decision, tool call, and order proposal.

Educational / research system. Not financial advice. AI can be wrong — verify every figure.

3. System Architecture

MBG is a layered multi-agent system orchestrated by Hermes. Specialist skills produce structured outputs; debate skills create adversarial or collaborative pressure; a Research Manager synthesizes; Risk and Portfolio layers gate; Shapley credit assignment updates blend weights from outcomes.

Layer	Responsibility
Orchestration (Hermes Master)	Schedule, route tasks, MEMORY, skill evolution, delivery (Telegram/CLI/PDF)
Data	OHLCV, fundamentals, news, on-chain; point-in-time; multi-source fallback
Analysis	Technical, Fundamental, Sentiment/News, Macro specialists
Debate & Research	Bull/Bear, collaborative, gated, regime-conditional, contest; Research Manager
Trading & Risk	Trader → Risk Manager → Portfolio Manager; conviction gating
Credit & Memory	Shapley assigner, outcome linking, strategist-memory
Backtest & Lab	Strategy simulation, research cards, walk-forward
Execution	Paper broker; supervised live adapters; kill-switch

Table 1 — Architectural layers.

4. Debate Strategies & Profit Orientation

Skill / Mode	Description	Prefer When
debate-adversarial-bull-bear	1–2 round Bull vs Bear + Research Manager	Ambiguous directional thesis
debate-safe-collaborative	Single-round Trust/Skeptic/Leader refinement	Report quality, multi-horizon
debate-consensus-gated	Debate then conviction/consensus threshold θ	High-noise (crypto), risk-first
debate-regime-conditional	Competitive vs collaborative by regime	Multi-regime IDX + crypto
debate-contest	Parallel paths scored after outcomes; re-allocate	Long-running continuous improvement
debate-orchestrator	Meta skill: selects mode, runs state machine, logs	Always the entry point

Table 2 — Debate strategy skills.

5. Shapley Value Agent Credit Assignment

Credit assignment answers: which agent or coalition actually contributed to profitable outcomes? The Shapley value is the unique fair division satisfying efficiency, symmetry, dummy, and additivity.

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \left[\frac{|S|! \cdot (|N| - |S| - 1)!}{|N|!} \cdot [v(S \cup \{i\}) - v(S)] \right]$$

For $N \leq 4$ specialists, exact enumeration of $2^N - 1$ coalitions is practical. $v(S)$ is defined from realized R-multiples or risk-adjusted contribution over an exponentially weighted window. Weights ω_i are normalized Shapley values, mixed with a Bayesian uniform prior during cold-start, and optionally scaled by regime multipliers. Inspired by Market Regime Council (MRC) results showing gains from Shapley-weighted integration rather than any single stage alone.

- Never use LLM self-confidence as $v(S)$ — only paper or live outcomes with costs.
- Persist CreditReport and causal trace for audit.
- Recompute on a schedule (e.g. weekly) once sufficient closed trades exist.

6. Hermes Skill Catalog (MBG)

All skills live under `~/hermes/skills/mbg/`. Each skill is a directory containing SKILL.md (and optional scripts/). Below are the concrete designs Hermes (or a developer) can implement directly.

6.1 Directory Layout

```
mbg/
■■■■ mbg-research-desk/ # Astra-style one-shot workflow
■■■■ debate-orchestrator/
■■■■ debate-adversarial-bull-bear/
■■■■ debate-safe-collaborative/
■■■■ debate-consensus-gated/
■■■■ debate-regime-conditional/
■■■■ debate-contest/
■■■■ research-manager/
■■■■ regime-detector/
■■■■ conviction-gater/
■■■■ risk-manager/
■■■■ shapley-credit-assigner/
■ ■■■■ scripts/shapley_exact.py
■■■■ strategist-memory/
■■■■ data-fetcher/
■■■■ technical-analyst/
■■■■ fundamental-analyst/
■■■■ sentiment-news-analyst/
■■■■ backtest-lab/
```

6.2 Embedded SKILL.md Designs (P0–P1)

A. mbg-research-desk (One-shot Research Desk)

name: mbg-research-desk

description: Run SCAN → ANALYZE → RESEARCH → RISK → PLAN for one ticker with hard safety rules

tags: [mbg, research, safety, one-shot]

When to Use

User asks for a full research plan on a single IDX or crypto symbol; quick deep-dive mode.

Procedure

1. Confirm inputs: ticker, market (IDX|CRYPTO), style, account size, max risk %, dated data or “fetch + cite”.
2. SCAN: is it worth researching now? If not, stop with reason.
3. ANALYZE: trend, momentum, support, resistance, price action — only from provided charts/data.
4. RESEARCH: FACTS table (metric|value|date|source); earnings; news CONFIRMED/RUMOR; catalysts. Bull/Bear cases.
5. RISK: stop from support; show position-size math; R:R vs nearest resistance (flag if <2:1); 3 invalidations; weakest assumption.
6. PLAN: fill one-page template (thesis, direction, entry, stop, targets, size, R:R, invalidation, key dates, chart vs research disagreement, confidence, numbers to verify).
7. End with STATUS: AWAITING HUMAN REVIEW.

Hard Rules: Never invent numbers. FACTS vs OPINION. No buy/sell language. Missing data = NOT PROVIDED — verify manually.

Verification: Plan contains arithmetic for size; R:R stated; STATUS line present; confidence labeled.

B. debate-orchestrator

description: Select debate mode, run state machine, persist transcript, return DebateResult

Procedure: Load analyst reports → call regime-detector → choose mode (adversarial default; gated for high noise; collaborative for stable; contest if history exists) → initialize debate_state → invoke chosen skill → log every turn to MEMORY → emit DebateResult JSON (mode, rounds, research_plan, transcript_ref, confidence).

Pitfalls: Do not default to high round counts (cost). Always store transcript_id for later Shapley joins.

C. debate-adversarial-bull-bear

inputs: analyst_reports[], symbol, market, max_rounds (default 1)

Procedure:

- Bull: strongest defensible long case; every claim cites analyst data.
- Bear: strongest short/avoid case; must rebut Bull's strongest point first.
- Alternate up to max_rounds.
- Research Manager: discrete rating (Buy/Overweight/Hold/Underweight/Sell or Long/Short/Neutral); prefer Hold when balanced; conviction 0–1; key risks.
- Return ResearchPlan + full history.

Prompt discipline: Force evidence anchors. Manager must not manufacture direction.

D. research-manager

Standalone adjudicator. Input: debate transcript + analyst reports. Output: structured ResearchPlan (rating, conviction, horizon, risks, evidence citations). Same discrete scale and Hold-when-balanced rule.

E. conviction-gater

Applies configurable θ (default 0.35–0.5) and portfolio constraints. If score $\geq \theta$ and R:R acceptable → emit GatedPlan; else HOLD with reason. Improves risk-adjusted outcomes by discarding low-conviction noise.

F. regime-detector

Classify regime for the relevant market: trend (bull/bear/range) + volatility (high/low). Features from price, volume, simple macro proxies (USDIDR, BI rate, crypto funding/Fear&Greed). Output: RegimeLabel + confidence + features used. Consumed by debate-orchestrator and

Shapley multipliers.

G. risk-manager

Protect capital. Inputs: ResearchPlan/TradingPlan draft, account size, max risk %, current price, chart levels.

1. Stop = level where idea is wrong (from support/structure) + why.
2. Position size math shown: $(\text{account} \times \text{risk}\%) \div (\text{entry} - \text{stop}) = \text{shares}$; \$ value and % of account.
3. R:R to nearest logical target; flag if under ~2:1.
4. Three invalidation conditions (level, event, news).
5. Weakest assumption.

Output: FACTS (math) → OPINION (Favorable/Even/Poor) + biggest risk. Always safety rules.

H. shapley-credit-assigner

When: Scheduled (weekly) or after batch of closed trades.

Procedure: Load exponential-decay performance histories → for $N \leq 4$ enumerate coalitions → define $v(S)$ from realized contribution → compute exact ϕ_i → normalize to ω_i → Bayesian mix with uniform prior → optional regime multipliers → write weights + CreditReport to MEMORY/config.

Verification: $\sum \omega_i \approx 1$; dummy agents near zero; human-readable report.

I. strategist-memory

Write/read decision logs, link closed trades to transcript_ids and debate modes, store user preferences and regime-mode profitability stats.

Critical path for Shapley: without outcome linkage, credit assignment cannot run.

7. End-to-End Pipelines

7.1 Daily / Scheduled Pipeline

1. Universe / watchlist update (LQ45, IDX30, user list, major crypto).
2. Parallel data fetch + specialist analysts.
3. regime-detector labels regime.
4. debate-orchestrator selects and runs debate mode(s).
5. research-manager (if needed) → ResearchPlan.
6. conviction-gater + risk-manager → GatedPlan or HOLD.
7. Trader produces TradingPlan; Portfolio Manager final gate.
8. Reporting skill publishes Daily Brief + Recommendation Pack (Markdown/PDF/Telegram).
9. After outcomes: strategist-memory links trades; shapley-credit-assigner updates weights on schedule.

7.2 Single-Ticker Deep Dive (mbg-research-desk)

User supplies ticker + data (or “fetch”). Skill runs SCAN→ANALYZE→RESEARCH→RISK→PLAN under full safety rules and stops at AWAITING HUMAN REVIEW. Can optionally call debate skills inside RESEARCH for higher rigor.

8. Trade Plan Schema & Human Template

Machine-readable JSON (for downstream skills) plus one-page human template. Minimum fields:

- symbol, market (IDX|CRYPTO), direction, conviction (0–1 or Low/Med/High)
- entry_zone / entry_condition, stop_loss (+ why), take_profit_tiers
- position_size (shares, \$, % account, \$ at risk) — arithmetic shown
- risk_reward, invalidation (list of 3), key_dates, weakest_assumption
- chart_vs_research_disagreement, confidence + what changes it
- numbers_to_verify, generated_at, status = AWAITING_HUMAN_REVIEW

Human one-page template (must be filled exactly):

```
TRADE PLAN — [TICKER] — [DATE]
THESIS · DIRECTION & TIMEFRAME · ENTRY · STOP (price + why) · TARGET 1 / 2
POSITION SIZE (shares, $, % account, $ at risk) · RISK/REWARD
INVALIDATION (3) · KEY DATES · CHART vs RESEARCH DISAGREEMENT
CONFIDENCE + what changes it · NUMBERS TO VERIFY
STATUS: AWAITING HUMAN REVIEW
```

9. Data Sources & Deployment

IDX: yfinance / local loaders for .JK tickers, Indonesian news (Detik, Kontan, CNBC Indonesia, X search), BI rate, USDIDR.

Crypto: CCXT / Binance / Bybit / CoinGecko, funding, OI, Fear & Greed.

All fetches must be dated and citable. Point-in-time correctness required for backtests.

Deployment targets: local laptop (Hermes CLI), Firebase (Functions + Firestore), Supabase (Edge + Postgres), Docker/VPS. Config via env + YAML (universe, risk limits, API keys, messaging tokens). Secrets never in source.

10. Implementation Roadmap

Phase	Focus	Deliverables
0 — Bootstrap	Hermes env + safety	mbg/ folder, safety rules in all prompts, data-fetcher stub, MEMORY layout
1 — Research Desk	One-shot + core debate	mbg-research-desk, debate-orchestrator, adversarial, research-manager, risk-manager
2 — Gating & Regime	Quality filters	conviction-gater, regime-detector, regime-conditional & consensus-gated
3 — Daily Pipeline	Continuous operation	Scheduled orchestrator, Daily Brief, paper portfolio tracking
4 — Credit & Learning	Shapley + memory	strategist-memory, shapley-credit-assigner, contest mode, weight-driven blend
5 — Hardening	Backtest + deploy	backtest-lab, Docker/Firebase/Supabase adapters, kill-switch, cost controls

Table 3 — Phased roadmap.

11. Success Metrics

- Safety: zero unauthorized live orders; every plan carries AWAITING HUMAN REVIEW until approved.
- Operational: daily pipeline success $\geq 95\%$; single-ticker research-desk completes with valid schema.
- Quality: user-rated usefulness of plans; hit rate of high-conviction gated calls over 30–90 days.
- Credit: Shapley weights stable after burn-in; dummy agents near zero; mode-tagged profitability tracked.
- Risk: no plan with R:R < 2:1 is marked Favorable without explicit warning.

12. Key Risks & Mitigations

Risk	Mitigation
LLM hallucination / invented numbers	Universal NOT PROVIDED rule; tool-first data; FACTS vs OPINION; human verify
Look-ahead in backtests	Point-in-time data layer; research-card checklist

Risk	Mitigation
Debate cost / latency	Cap rounds; cheap models for analysts; mode selection by need
Cold-start Shapley	Bayesian mixture with uniform prior until sample size sufficient
Over-trading / revenge	Conviction gating, daily loss breaker, human gate, kill-switch

Table 4 — Risks and mitigations.

Market Brain Grid (MBG) v1.2 — End of Product Requirements Document.
This document is the single source of truth for Hermes implementation. Install skills under `~/hermes/skills/mbg/`. Start with Phase 0–1 (safety + research-desk + adversarial debate). Every output remains research-grade until a human changes STATUS from AWAITING HUMAN REVIEW.

Not financial advice. Educational and research purposes only. Verify every number. Investing involves risk of loss.